ECONOMY · INDUSTRY · INNOVATION

TECHNOLOGICAL CHINA

From the world's factory to the innovation frontier

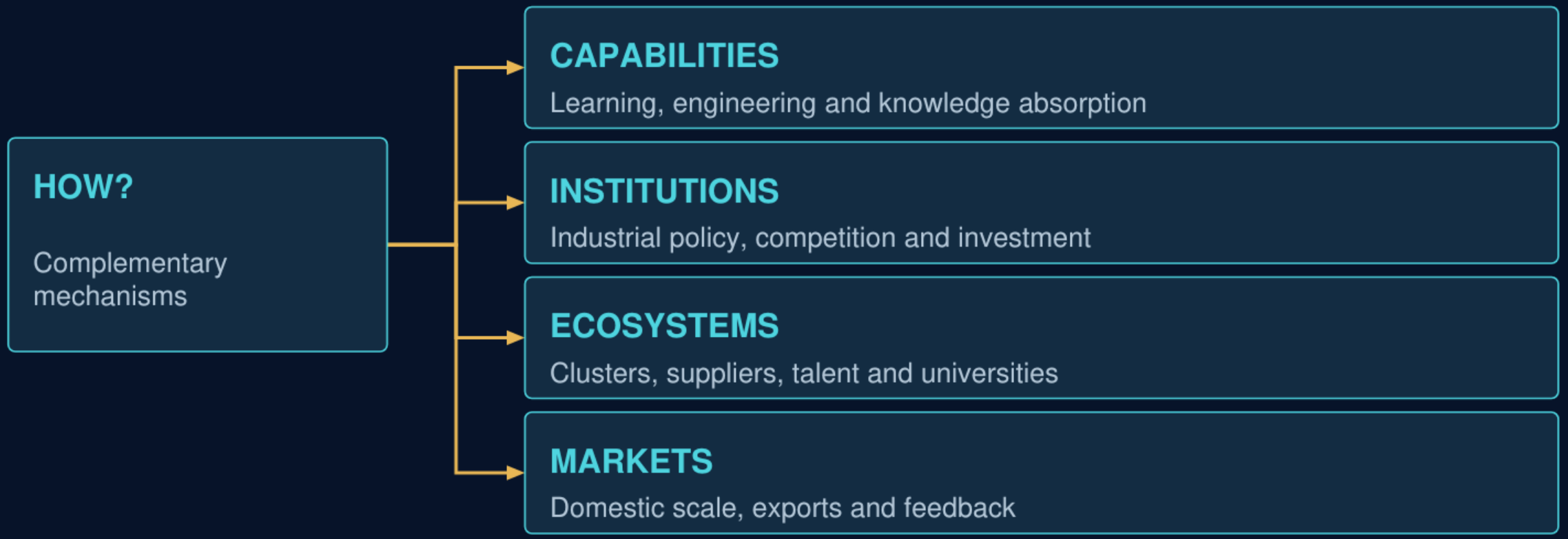
Sergio Gevatschnaider

HISTORICAL PERSPECTIVE · ECONOMIC ANALYSIS · TECHNOLOGICAL SYSTEMS



The central question

What explains the transition from assembly to domestic innovation?



Hypothesis: complementary mechanisms explain the transformation.

Five turning points in the transformation

From economic reform to competition in frontier technologies.



Reform and opening

Initial learning

Special zones

FDI and manufacturing

WTO accession

Export integration

Made in China 2025

Industrial upgrading

15th Five-Year Plan

Innovation and resilience

Reforms since 1978: building capabilities

Opening combined institutional experimentation with production integration.

Economic zones

Local experimentation, infrastructure and differentiated rules.

Foreign investment

Factories, management practices and market access.

Industrial learning

Process engineering, quality and suppliers.

Export scale

Production, productivity and competitive discipline.

Analytical point

Capability transfer requires learning, adaptation and supporting institutions. It is not automatic.

WTO 2001: integration and industrial scale

Accession deepened links with global supply chains.

2001

China's WTO accession

Markets

Trade predictability and broader access.

Investment

Links between firms and suppliers.

Scale

Expansion of manufacturing capacity.

Learning

Standards, quality and logistics.

External integration accelerated technological upgrading, without guaranteeing it.

Capturing value beyond product assembly

Producing at scale differs from controlling design, standards and knowledge.

Greater control of intangible assets and innovation rents

01
ASSEMBLY
Labour and logistics

02
PROCESSES
Quality, productivity

03
ENGINEERING
Design and adaptation

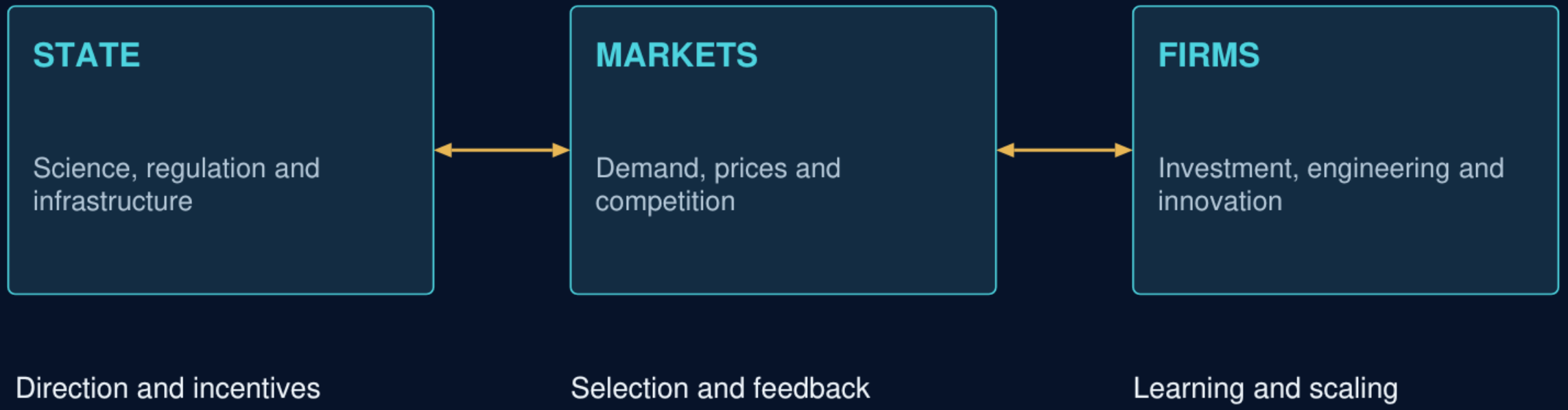
04
BRANDS
Networks and channels

05
R&D
Domestic technology

Conceptual ladder: sectors, firms and regions advance at different speeds.

China's industrial policy model

Public coordination interacts with market signals and business capabilities.



Reciprocal influence: coordination coexists with competition.

Made in China 2025

A manufacturing upgrading programme with multiple instruments and uncertain outcomes.

01 Information technology	02 Robotics and automation
03 Aerospace	04 Marine equipment
05 Advanced rail	06 New energy vehicles
07 Electrical equipment	08 Agricultural machinery
09 New materials	10 Biomedicine and devices

The 14th and 15th Five-Year Plans

Priorities extend digitalisation toward critical capabilities and deployment at scale.

CONTINUITY

14TH PLAN · 2021–2025

Technological innovation, digital economy growth, advanced manufacturing, industrial upgrading and infrastructure development.

DEEPENING

15TH PLAN · 2026–2030

Critical technologies, AI Plus applications, advanced chips, industrial software, basic research and integration of the real and digital economies.

Interpretive comparison: plans set priorities. Outcomes require evaluation.

“New quality productive forces”

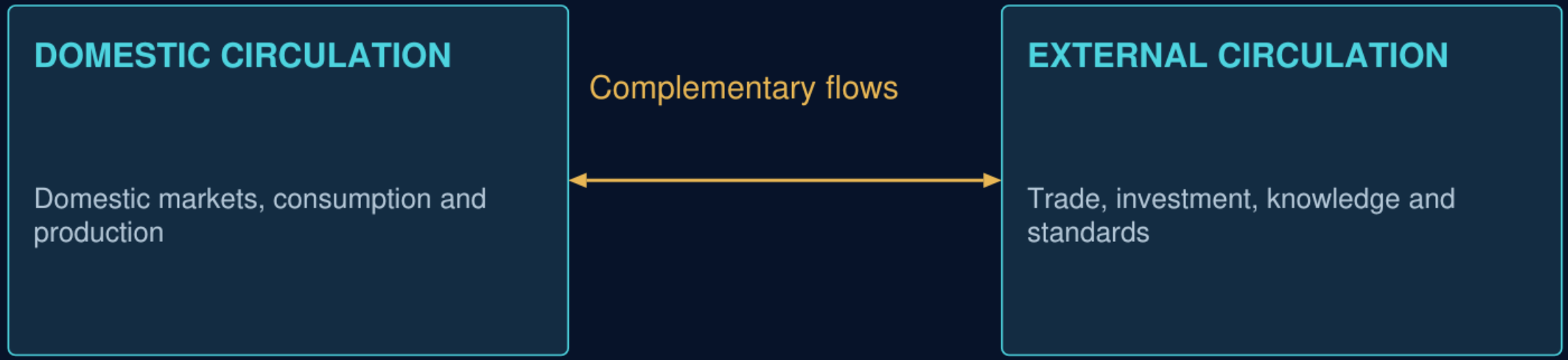
An economic policy concept: productivity gains through technological and organisational change.



Contributions depend on adoption, skills and institutions.

Dual circulation and technological resilience

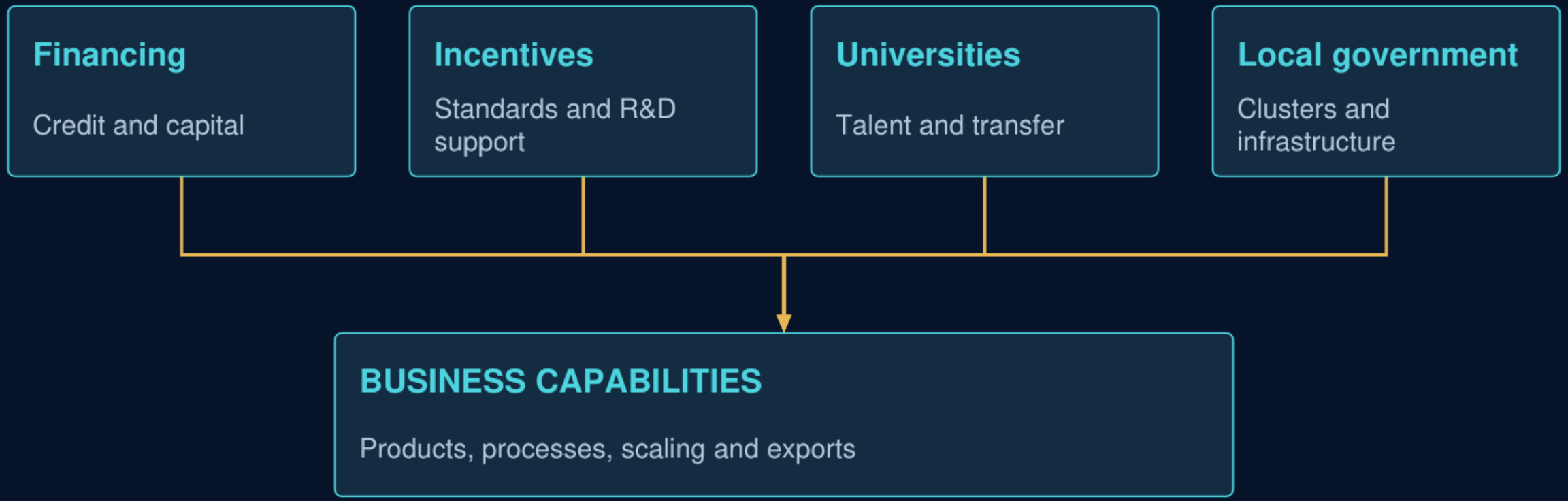
Domestic demand and external integration form complementary circuits.



Resilience alongside external integration: two interacting circuits.

Implementing the strategy

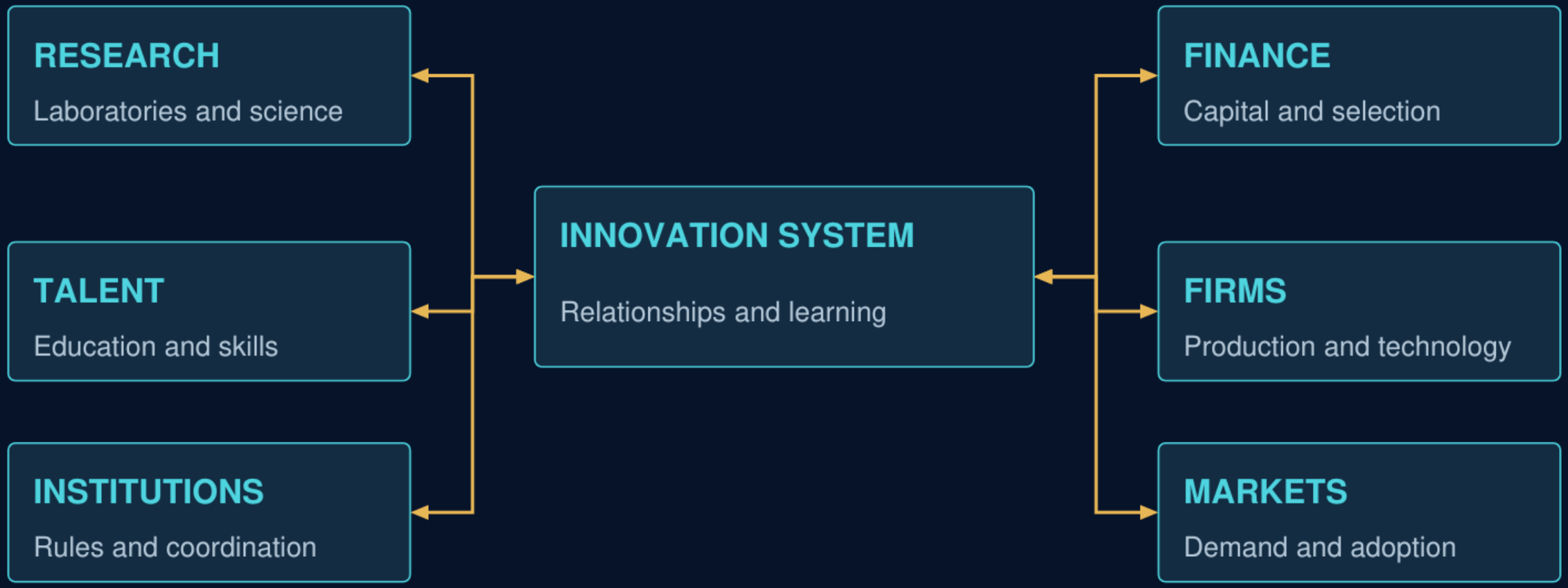
Innovation requires financing, knowledge and territorial coordination.



Risks: investment duplication, idle capacity and low social returns.

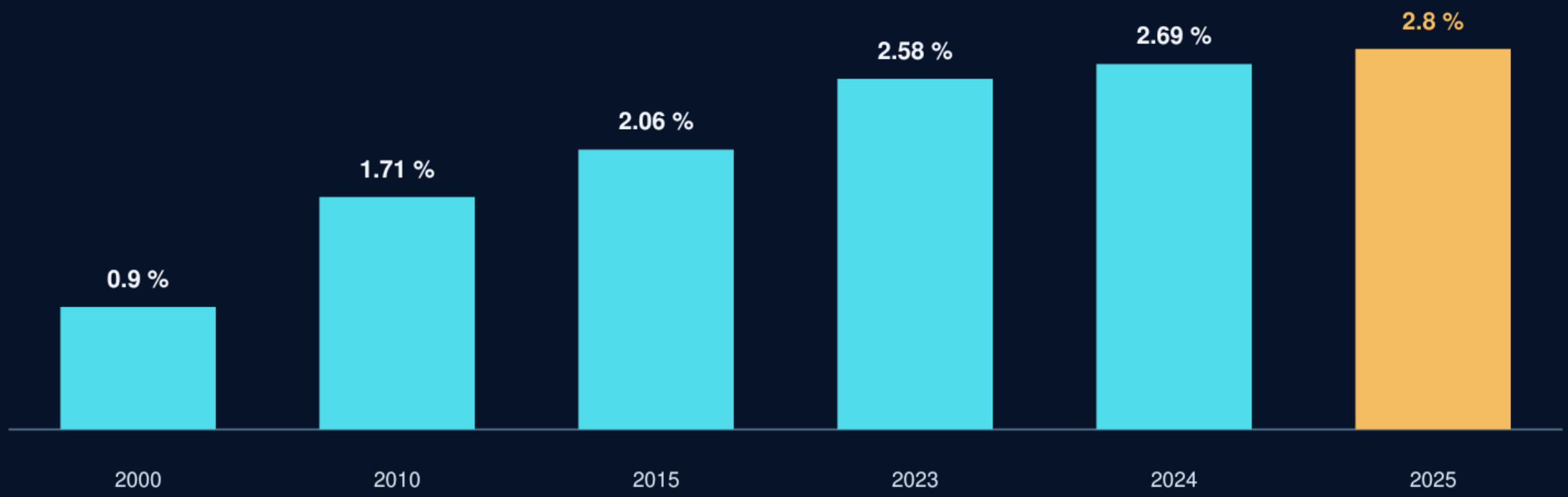
A national innovation system

A network that turns knowledge into products, processes and capabilities.



Sustained growth in R&D effort

Research and development intensity (% of GDP), selected years.



Historical series are subject to revision. 2025: consolidated official NBS data.

NBS China. World Bank / UNESCO series and methodological revisions.

R&D in 2025: scale and structure

Official data on research and development expenditure.

2.80 %
R&D as a share of GDP

3.93 tn
Yuan invested in R&D

77.4 %
Expenditure performed by firms

Composition by type of research



- Experimental development · 81.5%
- Applied research · 11.3%
- Basic research · 7.21%

NBS / MOST / Ministry of Finance. 2025 communiqué published in 2026.

Knowledge, talent and intellectual property

The innovation process connects research with adoption in real markets.

Knowledge

Basic and applied science, publications and laboratories.

Education

Engineers, scientists and technical staff.

Protection

Patents, standards and trade secrets.

Transfer

Spin-offs, licensing and collaboration.

Diffusion

Products, suppliers and users.

7,523

Huawei's published PCT applications in 2025

Patent counts require interpretation: quality, use and knowledge diffusion matter.

The geography of innovation: clusters in 2026

Inventors, research and finance concentrate in metropolitan ecosystems.

25

Chinese clusters in the global top 100

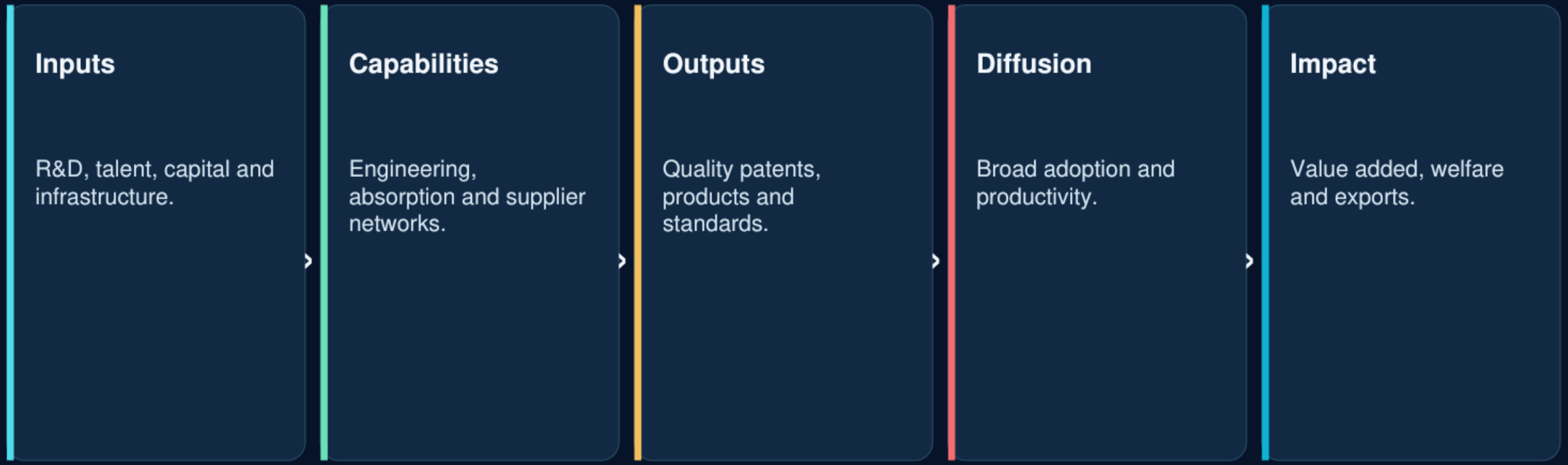
01	Shenzhen–Hong Kong–Guangzhou
04	Beijing
06	Shanghai–Suzhou
13	Hangzhou
32	Hefei

Shenzhen–Hong Kong–Guangzhou ranks first worldwide.

WIPO Global Innovation Index 2026. Corrected cluster ranking.

Measuring innovation beyond enthusiasm

A sound framework separates inputs, capabilities, outputs and impact.



Methodological caution: assess correlation, causality and selection bias separately.

A diverse business and technology ecosystem

Chinese technology firms differ in capabilities and business models.

HUAWEI

Networks · infrastructure · computing

ALIBABA

Commerce · cloud · AI

TENCENT

Digital services · ecosystems

BAIDU

Search · AI · mobility

BYD / CATL

Mobility and batteries

DJI / STARTUPS

Hardware · drones · innovation

Huawei: an interconnected technology system

A case of capability accumulation in research, infrastructure and software.



7,523

Published PCT applications in 2025

Interconnected capabilities. Global chip supply-chain dependencies persist.

Alibaba, Tencent and Baidu: platforms and capabilities

Data, digital infrastructure and AI connect large markets.

CLOUD + AI

ALIBABA

E-commerce, cloud and Qwen models.

ECOSYSTEMS

TENCENT

Messaging, games, payments and digital services.

AI + SERVICES

BAIDU

Search, AI, infrastructure and intelligent mobility.

Economic concepts: network effects, economies of scale, data and switching costs.

BYD, CATL and DJI: physical innovation and manufacturing

Product engineering and supplier integration support iterative improvement.

ELECTRIFIED VEHICLES

BYD

Component integration, manufacturing and distribution.

BATTERIES AND STORAGE

CATL

Electrochemistry, scaling and thermal management.

DRONES AND SENSORS

DJI

Hardware, software, control and user experience.

DeepSeek and Qwen: AI as a production layer

Available models expand applications. Infrastructure and organisation remain essential.

TECHNOLOGY LAYER

MODELS AND ALGORITHMS

Open or open-weight models, training and inference efficiency, reasoning, evaluation and tool use.

ECONOMIC LAYER

ADOPTION AND SYSTEMS

Business data, computing infrastructure, process integration, talent, agents and outcome measurement.

Model demonstrations alone do not establish measured productivity gains.

Shenzhen: the speed of industrial learning

A city as a laboratory linking hardware, software and markets.

Why does it matter?

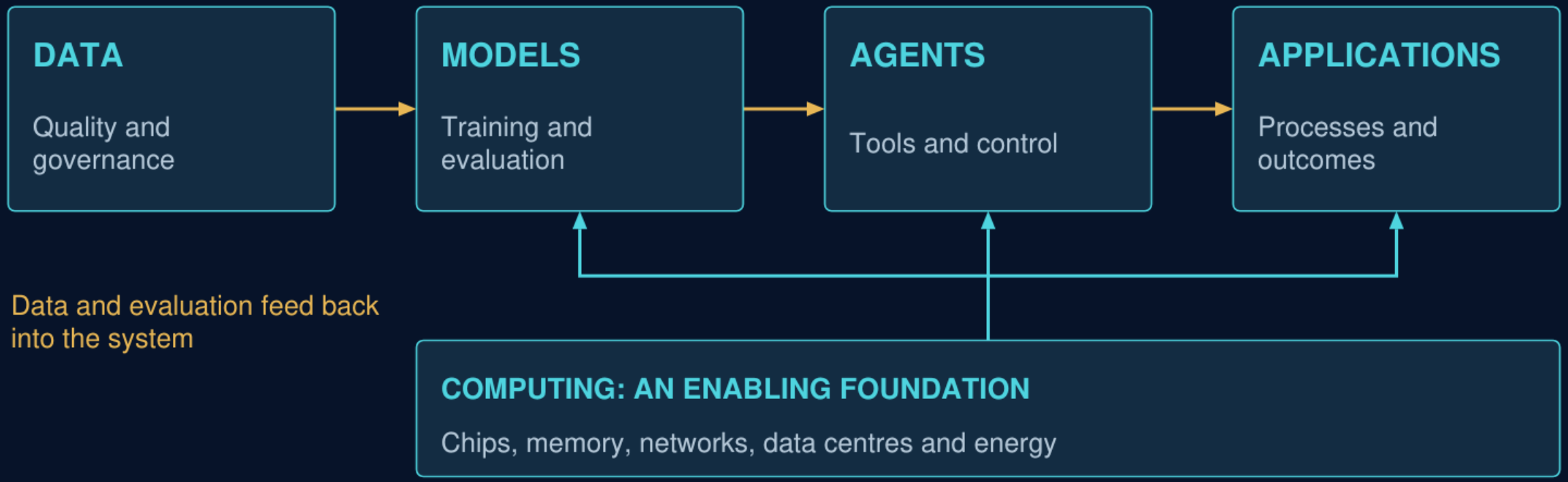
- Dense supplier and component networks.
- Rapid prototyping and redesign.
- Convergence of manufacturing and services.
- Talent, logistics and market access.



COMPLEMENTARY VIDEO · Shenzhen challenges Silicon Valley · FRANCE 24 Español

Industrial AI: models, infrastructure and processes

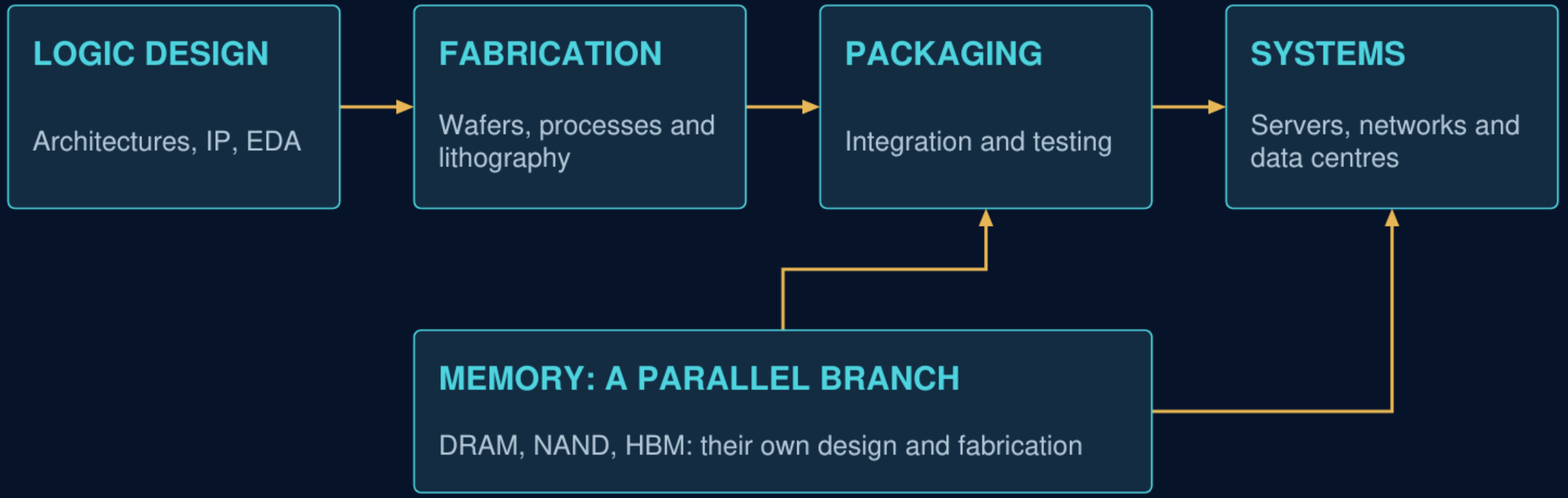
Interdependent layers of an industrial AI system.



Infrastructure enables training and inference across the architecture.

Semiconductors: the material basis of digital power

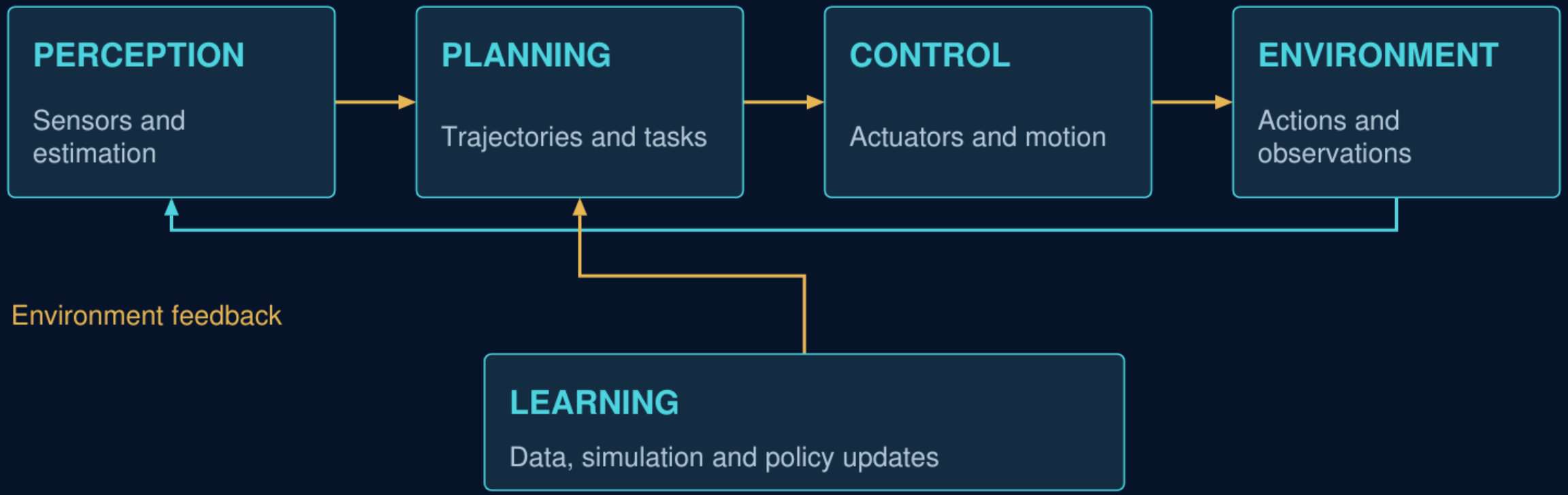
Application leadership can coexist with dependencies on critical components and equipment.



EDA, equipment, materials and software constrain different segments.

Robotics: AI acting in the physical world

Perception, planning and control extend programmed industrial automation.



Learning updates policies. The operational loop acts in real time.

Electrification: production, batteries and learning

China has become a central hub of the global electric car industry.

≈55%

of new car sales in China (2025)

≈75%

of global electric car production (2025)

Batteries
Materials and electrochemistry

→

Vehicles
Design, software and assembly

→

Grid and charging
Electricity and infrastructure

→

Markets
Demand and competition

IEA: electric cars include battery electric and plug-in hybrid cars.

Multiple technological frontiers

Advanced technology extends beyond AI, chips and robots.

QUANTUM COMPUTING

Research and demonstrations. Scaling remains open.

BIOTECHNOLOGY

Biomedical R&D, materials and biomanufacturing.

SPACE

Satellites, navigation and associated services.

ADVANCED NETWORKS

Connectivity, photonics and distributed systems.

Distinguish promising research, prototypes, commercial production and widespread adoption.

China and the United States: competition and interdependence

Technology competition affects components, markets, standards and security.

TECHNOLOGICAL RIVALRY

AREAS OF COMPETITION

Advanced chips, digital infrastructure, AI, research, standards, talent and market access.

Regulation and export controls reshape value chains.

COOPERATION / EXCHANGE

PERSISTENT INTERDEPENDENCIES

Bilateral trade, research, manufacturing, global suppliers and consumers.

Diversification can increase costs in the short term.

Which segments are being reconfigured, and at what cost?

System strengths and vulnerabilities

Industrial scale coexists with technological dependencies and efficiency challenges.

STRENGTHS

- Large manufacturing and supplier capacity.
- Domestic market, infrastructure and scale.
- Regional innovation ecosystems.
- Rapid adoption across sectors.

VULNERABILITIES

- Imported critical components and equipment.
- Gaps in specialised capabilities.
- Capital allocation and excess capacity.
- Trade and demographic risks.

Technological autonomy and dependency vary across sectors.

Domestic challenges to technological growth

Innovation operates within an economy that needs to rebalance its development model.

01	**PRODUCTIVITY**	Investment must improve efficiency as well as volume.
02	**DOMESTIC DEMAND**	Consumption, services and household expectations.
03	**DEMOGRAPHICS**	Ageing and changes in the workforce.
04	**REAL ESTATE**	Sectoral adjustment and links to local finances.
05	**CAPITAL**	Discipline, returns and resource allocation.

Implications for Latin America

Trade opportunities coexist with industrial competition and concentration risks.

TRADE

Exports, imports and sectoral change.

INFRASTRUCTURE

Energy, transport, digitalisation and finance.

VALUE CHAINS

Minerals, agribusiness, services and components.

LOCAL CAPABILITIES

Training, standards, R&D and supplier linkages.

Policy question: how can countries capture more local value while retaining technology and market access?

Five conclusions on technological China

Development economics, engineering, innovation and geoeconomics explain the transformation.

01	**SCALE**	Manufacturing builds capabilities and markets. Frontier leadership requires more.
02	**LEARNING**	Innovation accumulates within firms, teams and supplier networks.
03	**COORDINATION**	State and market interaction brings benefits and risks.
04	**COMPETITION**	Domestic and external pressures change incentives and strategies.
05	**INTERDEPENDENCE**	Autonomy varies by sector. Global supply chains remain important.

THE CENTRAL QUESTION

Can China turn industrial scale into sustained innovation?



Production capability ≠ commercial leadership ≠ technological autonomy

Sergio Gevatschnaider

KEY SOURCES: UNIDO · WIPO · NBS · IEA · WORLD BANK